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Mg^{2+}-promoted high-efficiency DNA conjugation on polydopamine surfaces for aptamer-based ochratoxin A detection

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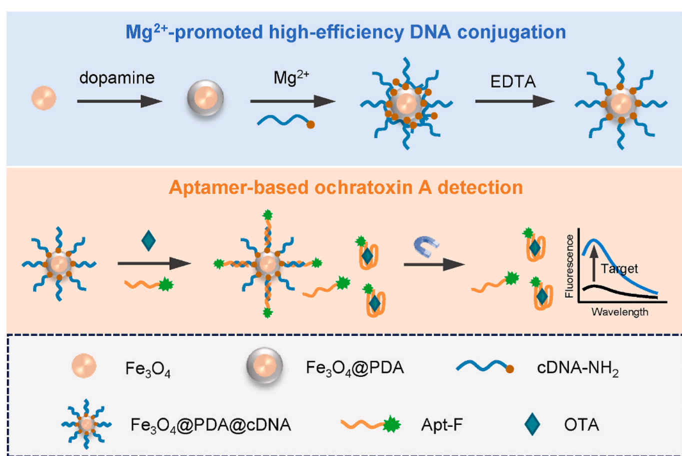
HIGHLIGHTS

- Mg^{2+}/Ca^{2+}-promoted efficient DNA conjugation on polydopamine-modified surfaces.
- DNA immobilization on PDA-coated magnetic nanoparticles and PDA-coated 96-well plates.
- Fe_3O_4@PDA@cDNA nanoparticles with high DNA hybridization capacity and minimal non-specific adsorption.
- Utilization of Fe_3O_4@PDA@cDNA nanoparticles and FAM-labelled aptamers for ochratoxin A detection.

GRAPHICAL ABSTRACT



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ABSTRACT

Background: Surface immobilization of DNA is the foundation of a broad range of applications in biosensing and specific DNA extraction. Polydopamine (PDA) coatings can serve as intermediate layers to immobilize amino- or thiol-labelled molecules, including DNA, onto various materials through Michael addition and/or Schiff base reactions. However, the conjugation efficiency is limited by electrostatic repulsion between negatively charged DNA and PDA. Recently, it has been reported that polyvalent metal ions (such as Mg^{2+} and Ca^{2+}) can mediate the adsorption of DNA on PDA surfaces. Inspired by this, in this work we aimed to exploit polyvalent metal ions to facilitate the conjugation of DNA on PDA.

Results: Mg^{2+} was used to promote the conjugation of amino-terminated DNA complementary to ochratoxin A (OTA) aptamer (cDNA-NH_2) on PDA-coated magnetic nanoparticles (Fe_3O_4@PDA). After the reaction, the unlinked cDNA-NH_2 adsorbed on Fe_3O_4@PDA mediated by Mg^{2+} was removed with EDTA. In the presence of 20 mM Mg^{2+}, the amount of covalently linked cDNA-NH_2 increased approximately 11-fold compared to that in the absence of Mg^{2+}. The resulting Fe_3O_4@PDA@cDNA conjugates exhibited superior hybridization capacity towards OTA aptamers, minimal nonspecific adsorption, and excellent chemical stability. The conjugates combined

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with fluorophore-labelled aptamers were employed for OTA detection, achieving a limit of detection (LOD) of 2.77 ng mL^{-1}. To demonstrate versatility, this conjugation method was extended to Ca^{2+}-promoted conjugation of cDNA-NH_2 on Fe_3O_4@PDA nanoparticles and Mg^{2+}-promoted conjugation of cDNA-NH_2 on PDA-coated 96-well plates.

Significance: The conjugation efficiency of DNA on PDA was significantly improved with the assistance of polyvalent metal ions (Mg^{2+} and Ca^{2+}), providing a facile and efficient method for DNA immobilization. Due to the substrate-independent adhesion property of PDA, this method demonstrates versatility in DNA surface modification and holds great potential for applications in target extraction, biosensing, and other fields.

1. Introduction

The immobilization of oligonucleotides, including aptamers, on material surfaces is crucial for various applications, such as biosensing, bioimaging, and target extraction [1,2]. The frequently used approaches for immobilizing end-functionalized DNA include amino-terminated DNA immobilized on carboxylated surfaces via 1-ethyl-3-(3-dimethylaminopropyl) carbodiimide/N-hydroxysuccinimide (EDC/NHS) chemistry; amino-terminated DNA attached to amino-modified surfaces using glutaraldehyde as a crosslinker; biotinylated DNA attached to avidin- or streptavidin-modified surfaces through affinity interactions; thiol-labelled DNA self-assembled on gold surfaces via Au–S bonds; and click reaction of azide–alkyne cycloadditions [3,4]. The selection of the immobilization method is dependent on the surface chemistry of the materials used. When evaluating a conjugation approach, important factors include chemical stability, conjugation efficiency, target capturing ability, nonspecific adsorption, and others. The conjugation chemistry used for DNA immobilization is closely linked to the assay performance. Notably, certain conjugation methods may result in a higher degree of nonspecific adsorption [5,6], or multiple modification steps are needed prior to DNA immobilization [7]. In the case of Au–S bond-based conjugation, the immobilized DNA strands are vulnerable to replacement by other thiol-containing molecules [8].

Polydopamine (PDA) can be deposited on a wide range of substrates through the spontaneous polymerization of dopamine under alkaline conditions. The resulting PDA coatings can react with thiol- or amine-containing molecules, including DNA, through Michael addition and/or Schiff base reactions [9,10]. This provides a facile and universal method for DNA immobilization on various substrates by using PDA as an intermediate layer. The attachment of DNA strands on PDA-modified nanomaterials or electrode surfaces has been employed to develop diverse optical and electrochemical biosensors [11,12]. As a biocompatible material, PDA can protect the attached DNA from nuclease degradation, making it suitable for applications in nanomedicine and intracellular imaging [13,14].

However, the conjugation of DNA to PDA is inefficient due to electrostatic repulsion between negatively charged DNA and PDA [1]. To overcome this limitation and enhance the conjugation efficiency, a protocol in which NaCl was added at a typical concentration of 150 mM has been reported [14–16], similar to the salt-ageing method used for the self-assembly of thiol-modified DNA on gold surfaces [17]. It was recently reported that polyvalent metal ions, such as Mg^{2+}, Ca^{2+}, and Zn^{2+}, could facilitate the adsorption of DNA on the PDA surface, whereas monovalent metal ions such as Na^+ and K^+ could not [18,19]. Inspired by this, we supposed that the introduction of polyvalent metal ions might further improve the conjugation efficiency due to the increased reaction probability assisted by polyvalent metal-mediated DNA adsorption.

In this study, we used Mg^{2+} to promote the conjugation of amino-modified DNA complementary to ochratoxin A aptamers (cDNA-NH_2) on magnetic Fe_3O_4@PDA core-shell nanoparticles. Ochratoxin A (OTA) is a common mycotoxin and is classified as a Group 2B carcinogen. The conditions of Mg^{2+} concentration, cDNA-NH_2 dosage, and reaction time were optimized. After the reaction, the unlinked cDNA-NH_2 adsorbed on Fe_3O_4@PDA mediated by Mg^{2+} was removed through EDTA-Mg^{2+} chelation. The obtained Fe_3O_4@PDA@cDNA conjugates exhibited excellent chemical stability, minimal nonspecific adsorption, high hybridization capacity, and reusability. The conjugates together with FAM-labelled OTA aptamers (Apt-F) were utilized for OTA detection, achieving a limit of detection (LOD) of 2.77 ng mL^{-1}. Real samples of OTA-spiked red wines were analysed via this method. Finally, to demonstrate versatility, Ca^{2+}-promoted cDNA-NH_2 conjugation on Fe_3O_4@PDA nanoparticles and Mg^{2+}-promoted cDNA-NH_2 conjugation on a PDA-coated 96-well plate were performed successfully.

2. Experimental section

2.1. Reagents and apparatus

$FeCl_3 \cdot 6H_2O$, anhydrous sodium acetate, trisodium citrate dihydrate, and ethylene glycol were obtained from Kermel (Tianjin, China). Dopamine hydrochloride was purchased from J&K Chemicals (Beijing, China). OTA, aflatoxin B1 (AFB1), zearalenone (ZEN), deoxynivalenol (DON), and T-2 toxin (T-2) were obtained from Pribolab (Singapore). An OTA enzyme-linked immunosorbent assay (ELISA) kit was purchased from Meimian (Jiangsu, China). The oligonucleotides used in this study (listed in Table S1) were purchased from Sangon (Shanghai, China). Super Gold nucleic acid dye (10000 ×) was obtained from Fluorescence (Beijing, China). NaCl and ethylenediamine tetraacetic acid (EDTA) were obtained from Aladdin (Shanghai, China). $MgCl_2 \cdot 6H_2O$ was obtained from Meryer (Shanghai, China). $CaCl_2 \cdot 2H_2O$ was obtained from Xiya (Shandong, China). Tris(hydroxymethyl)aminomethane (Tris) was obtained from Acmec (Shanghai, China). The red wine was purchased from a local supermarket.

Transmission electron microscopy (TEM) images were acquired with a JEM-2100 transmission electron microscope (JEOL, Japan). The ζ-potential measurements were carried out on a Malvern Zetasizer Nano ZS90. X-ray photoelectron spectroscopy (XPS) was performed on a Thermo Fisher ESCALAB 250XI instrument with Al Kα (1486.6 eV) as the X-ray source. Magnetization curves were measured with a vibrating sample magnetometer (VSM, LakeShore 7404, America) at room temperature. The fluorescence spectra were recorded on a Hitachi F-4600 spectrophotometer.

2.2. Synthesis of the Fe_3O_4@PDA core-shell nanoparticles

First, magnetic Fe_3O_4 cores were synthesized through a solvothermal reaction following the method reported by Zhao et al. [20]. Specifically, 1.08 g of $FeCl_3$ $6H_2O$, 0.40 g of trisodium citrate dihydrate, and 1.20 g of sodium acetate were dissolved in 20 mL of ethylene glycol under stirring. The resulting mixture was transferred to a Teflon-lined stainless-steel autoclave and sealed before being heated at 200 °C for 10 h. After cooling to room temperature, the obtained Fe_3O_4 magnetic nanoparticles were thoroughly washed with ethanol and deionized water several times before being dried in a vacuum oven.

For the formation of PDA shells, 3 mg of Fe_3O_4 nanoparticles was dispersed in 24 mL of Tris-HCl buffer (10 mM, pH 8.5) via ultrasonication. Subsequently, 7 mg of dopamine hydrochloride was added, and the mixture was allowed to react for 5 h under shaking. After magnetic separation, the collected Fe_3O_4@PDA nanoparticles were redispersed in Tris-HCl buffer and shaken for an additional 4 h to remove loosely adsorbed dopamine and promote further polymerization of the coated PDA shells. Finally, the Fe_3O_4@PDA core-shell nanoparticles were dispersed in 3 mL of Tris-HCl buffer and stored at 4 °C for future use (considered 1 mg mL^{-1}).

2.3. Mg^{2+}-promoted conjugation of cDNA-NH_2 on Fe_3O_4@PDA nanoparticles

cDNA-NH_2 molecules (1–5 nmol) were mixed with Fe_3O_4@PDA nanoparticles (1 mg) in 1 mL of Tris-HCl buffer (10 mM, pH 8.5) containing 150 mM NaCl and 0–20 mM $MgCl_2$. The mixtures were reacted at room temperature for 2–10 h under shaking. Covalent attachment of cDNA-NH_2 onto Fe_3O_4@PDA was achieved via Michael addition and/or Schiff base reactions. The resulting Fe_3O_4@PDA@cDNA conjugates were washed three times with a TE solution (10 mM Tris-HCl, 5 mM EDTA, pH 8.5) for 20 min each time to remove unreacted cDNA-NH_2. Subsequently, the Fe_3O_4@PDA@cDNA conjugates were dispersed in 1 mL of Tris-HCl buffer and stored at 4 °C for future use (considered 1 mg mL^{-1}). The reaction supernatants and TE washing solutions were collected and analysed to quantify the amount of the attached cDNA-NH_2.

2.4. Mg^{2+}-promoted conjugation of cDNA-NH_2 on PDA-coated 96-well plates

Dopamine hydrochloride was dissolved in Tris-HCl buffer (10 mM, pH 8.5) at a concentration of 2 mg mL^{-1}. Subsequently, 200 μL of the freshly prepared dopamine solution was added to each well and allowed to coat the cells overnight. The PDA-coated wells were washed at least three times with Tris-HCl buffer. For cDNA-NH_2 conjugation, 200 μL of cDNA-NH_2 (1 μM) in conjugation buffer (10 mM Tris-HCl, 150 mM NaCl, 20 mM $MgCl_2$, pH 8.5) was added to each PDA-coated well and incubated for 10 h. Then, the unreacted cDNA-NH_2 molecules were removed by washing three times with a TE solution.

2.5. Aptamer-based OTA detection

Fe_3O_4@PDA@T10cDNA (cDNA-NH_2 with a T10 spacer) conjugates and FAM-labelled OTA aptamers (Apt-F) were used for OTA detection. Specifically, a mixture of 300 μL of binding buffer (10 mM Tris-HCl, 1 M NaCl, 6 mM $CaCl_2$, 0.02% Tween 20, pH 8.5), 50 μL of Apt-F (500 nM), and 75 μL of varying concentrations of OTA was incubated for 5 min. Then, 75 μL of Fe_3O_4@PDA@T10cDNA (1 mg mL^{-1}) was added, and the mixture was further incubated for 20 min to capture uncombined Apt-F. After magnetic separation, the supernatants were collected for fluorescence measurements.

3. Results and discussion

3.1. Synthesis and characterization of the Fe_3O_4, Fe_3O_4@PDA, and Fe_3O_4@PDA@cDNA nanoparticles

Magnetic nanoparticles have been widely applied in target separation, enrichment and sensing due to their unique magnetic properties [21]. In this study, highly water-dispersible Fe_3O_4 magnetic nanoparticles were synthesized via a solvothermal reaction using citrate as a stabilizer [20]. As shown in Fig. 1A, the obtained Fe_3O_4 nanoparticles exhibited excellent dispersity and nearly spherical shapes with an average diameter of 122 nm. Following 5 h of modification, PDA shells with a thickness of ~30 nm were formed around the Fe_3O_4 cores (Fig. 1B). The thickness of the PDA shells can be adjusted by varying the concentration of dopamine or reaction time [22]. Some degree of aggregation was observed after PDA coating. Both the Fe_3O_4 and Fe_3O_4@PDA nanoparticles displayed superparamagnetic properties as evidenced by the absence of hysteresis in the magnetization curves (Fig. 1C). After the formation of the PDA shells, the saturation magnetization decreased from 47.9 emu g^{-1} to 29.8 emu g^{-1}.

The XPS survey spectra (Fig. 1D) revealed that the Fe 2p peaks at 724.35 eV and 710.81 eV disappeared, the C 1s peak at 285.07 eV increased, and the N 1s peak at 399.87 eV emerged after PDA modification [23]. Moreover, the nitrogen-to-carbon (N/C) ratio is 0.131 for Fe_3O_4@PDA, which is close to the theoretical value of dopamine

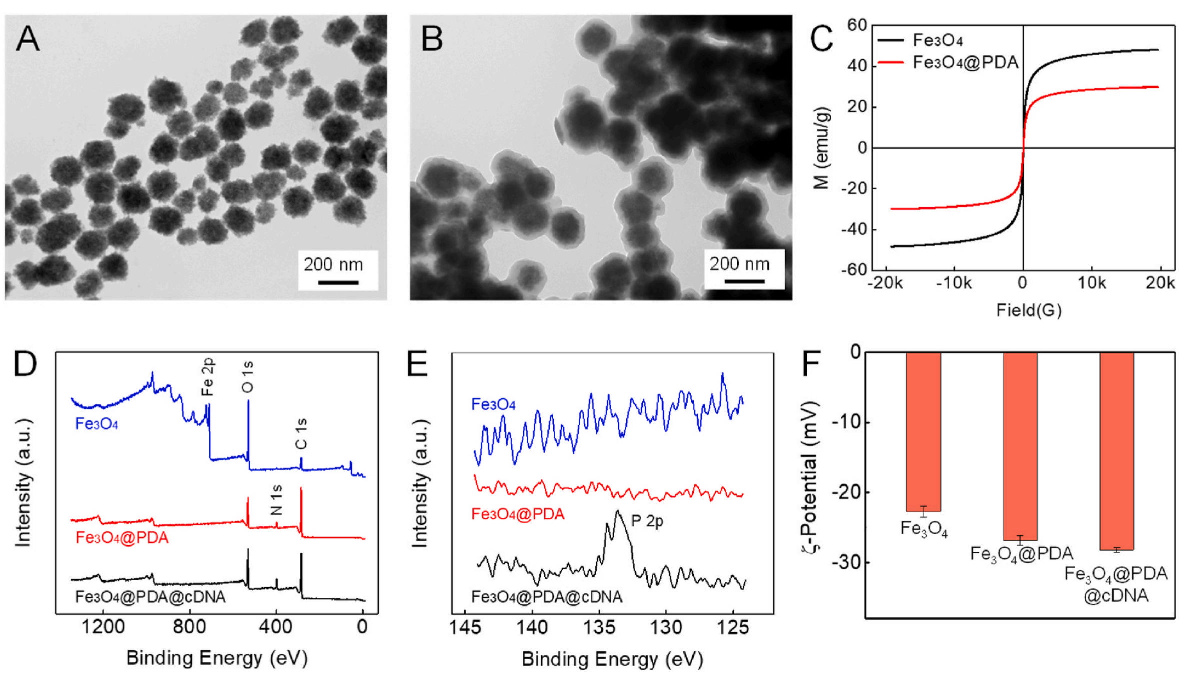

Fig. 1. TEM micrographs of the Fe_3O_4 (A) and Fe_3O_4@PDA (B) nanoparticles. (C) Room-temperature magnetization curves of the Fe_3O_4 and Fe_3O_4@PDA nanoparticles. XPS survey spectra (D) and high-performance P 2p spectra (E) of the Fe_3O_4, Fe_3O_4@PDA, and Fe_3O_4@PDA@cDNA nanoparticles. (F) ζ-potentials of the Fe_3O_4, Fe_3O_4@PDA, and Fe_3O_4@PDA@cDNA nanoparticles measured in 10 mM Tris-HCl at pH 7.0.

(0.125), confirming the formation of PDA shells and complete coverage of the Fe_3O_4 cores [9]. After cDNA-NH_2 immobilization, the N 1s and O 1s peaks at 400.15 eV and 532.71 eV, respectively, increased, while the C 1s peak at 285.63 eV decreased (Fig. 1D). Additionally, a characteristic peak corresponding to DNA, P 2p at 133.75 eV, emerged (Fig. 1E), indicating that cDNA-NH_2 was linked successfully [24]. The ζ-potentials of the Fe_3O_4, Fe_3O_4@PDA, and Fe_3O_4@PDA@cDNA were measured in Tris-HCl (10 mM, pH 7.0). All the nanoparticles were negatively charged, and their ζ-potentials decreased from −22.7 mV to −26.9 mV and −28.2 mV after PDA modification and cDNA-NH_2 conjugation, respectively (Fig. 1F).

3.2. *Mg^{2+}-promoted conjugation of cDNA-NH_2 on Fe_3O_4@PDA nanoparticles*

PDA is commonly used as an intermediate layer to functionalize material surfaces by reacting with amino- or thiol-containing molecules [25,26]. However, it is difficult for DNA molecules to access PDA surfaces due to charge repulsion, resulting in a low conjugation efficiency. It was previously reported that polyvalent metal ions could mediate DNA adsorption on PDA surfaces [18,19]. Accordingly, we hypothesized that this process may facilitate the covalent conjugation of DNA on PDA. To test this concept, amino-labelled DNA (cDNA-NH_2), which is partially complementary to the OTA aptamer, was conjugated with PDA-coated Fe_3O_4 nanoparticles in the presence of Mg^{2+}. Amino-labelled DNAs were chosen instead of thiol-labelled DNAs because thiol groups are susceptible to oxidation and require pretreatment to cleave disulfide bonds [27]. In a previous study by Duan et al. [14], NaCl was added at a concentration of 150 mM to facilitate covalent linkage between DNA and PDA. Building on this, we additionally introduced Mg^{2+} to further increase the conjugation efficiency. After the conjugation reaction, unreacted cDNA-NH_2 molecules adsorbed on Fe_3O_4@PDA were removed by chelation of EDTA and Mg^{2+}. This process of Mg^{2+}-promoted conjugation of cDNA-NH_2 on Fe_3O_4@PDA is illustrated in Fig. 2A.

To verify the promoting effect of Mg^{2+}, cDNA-NH_2 (2 nmol) was reacted with Fe_3O_4@PDA (1 mg) in 1 mL of Tris-HCl buffer (10 mM, pH 8.5) containing 150 mM NaCl and either 0 or 20 mM $MgCl_2$. After a duration of 10 h, the magnetic particles were collected and washed three times with TE solution. The amount of covalently linked cDNA-NH_2 was quantified by analysing the reaction supernatants and washing solutions. The nucleic acid dye Super Gold was utilized for DNA quantification [28]. The fluorescence spectra of the reaction supernatants and first-wash TE solutions stained with Super Gold are shown in Fig. S1. Obviously, more cDNA-NH_2 molecules were retained in the reaction supernatant in the absence of Mg^{2+}, while more cDNA-NH_2 molecules were washed off in the TE solution in the presence of 20 mM Mg^{2+}. The linked cDNA-NH_2 molecules were quantified as ~0.14 and ~1.59 nmol for 0 and 20 mM Mg^{2+}, respectively, exhibiting an 11-fold increase (Fig. 2B). These results proved that Mg^{2+} strongly promoted the conjugation of cDNA-NH_2 onto PDA surfaces. In addition, EDTA effectively eliminated the unreacted cDNA-NH_2 adsorbed on PDA.

The Fe_3O_4@PDA@cDNA conjugates prepared with and without Mg^{2+} were compared by hybridization with FAM-labelled OTA aptamers (Apt-F). Both Fe_3O_4@PDA@cDNA conjugates (0.1 mg mL^{-1}) were separately incubated with 50 nM Apt-F for 60 min, after which the supernatants were subjected to fluorescence measurements after magnetic separation (Fig. S2). Obviously, the conjugates prepared with Mg^{2+} had a greater hybridization capacity, with ~80% and ~9% of Apt-F hybridized on the Fe_3O_4@PDA@cDNA conjugates, corresponding to loading capacities of ~0.40 and ~0.047 nmol Apt-F per milligram of

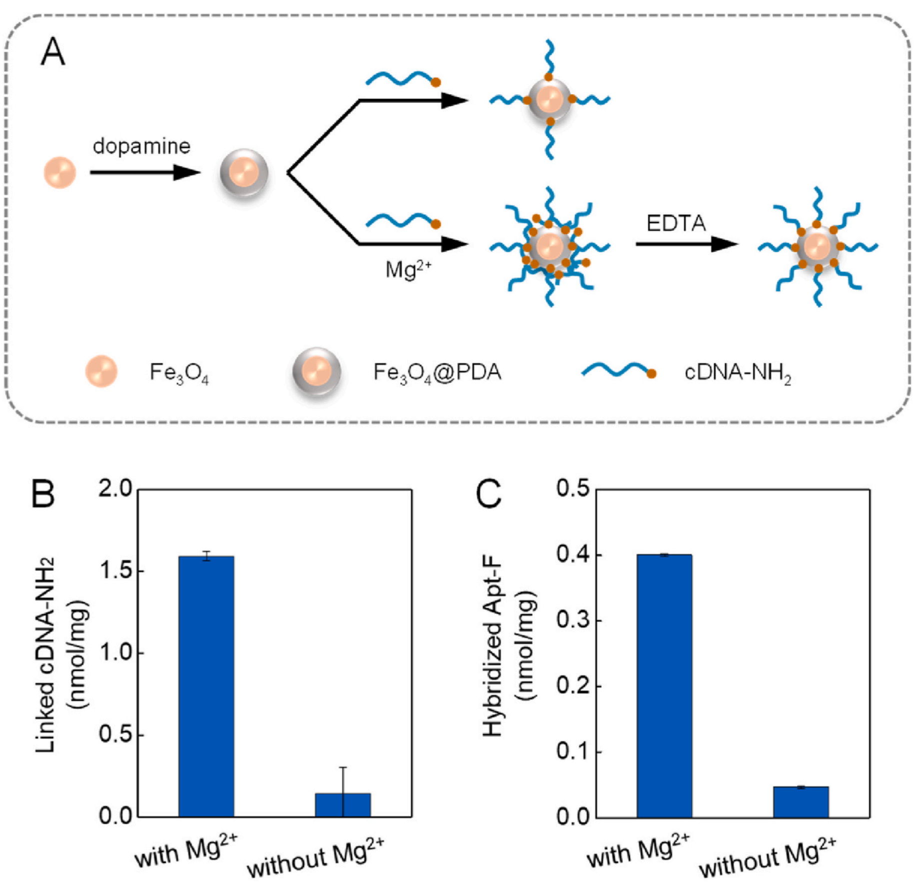

Fig. 2. (A) A schematic diagram of the Mg^{2+}-promoted conjugation of cDNA-NH_2 on Fe_3O_4@PDA nanoparticles. (B) The linked amounts of cDNA-NH_2 per milligram of Fe_3O_4@PDA under conditions either with or without 20 mM Mg^{2+}. (C) The hybridization capacities of the obtained Fe_3O_4@PDA@cDNA conjugates towards Apt-F.

conjugates, respectively (Fig. 2C). We also conducted a conjugation experiment in the presence of 20 mM Mg^{2+} without 150 mM NaCl, which yielded comparable results in terms of the linked amount of cDNA-NH_2 (~1.73 nmol) and hybridization percentage (~82%) (Fig. S3). However, the conjugation experiments in this study were performed in the presence of 150 mM NaCl.

3.3. Optimization of the conjugation conditions

To achieve a high conjugation density, the reaction conditions, including the Mg^{2+} concentration, cDNA-NH_2 dose, and reaction time, were optimized. For 1 mg of Fe_3O_4@PDA dispersed in 1 mL of conjugation buffer, Mg^{2+} concentrations ranging from 0 to 20 mM, cDNA-NH_2 dosages ranging from 1 to 5 nmol, and reaction times ranging from 2 to 10 h were investigated individually. The amount of linked cDNA-NH_2 at each test condition was quantified using the nucleic acid dye method as discussed above. The results are plotted in Fig. 3A–C. The hybridization capacities of the obtained Fe_3O_4@PDA@cDNA conjugates (0.1 mg mL^{-1}) were studied by incubating them with Apt-F (50 nM). The calculated hybridization percentages of Apt-F are shown in Fig. 3D–F. These results demonstrated that increasing the Mg^{2+} concentration, increasing the cDNA-NH_2 dose, or prolonging the reaction time led to a greater conjugation density and hybridization capacity. Ultimately, 20 mM Mg^{2+}, 2 nmol of cDNA-NH_2, and 10 h of reaction time were used for DNA conjugation.

3.4. Key properties of the Fe_3O_4@PDA@cDNA conjugates

The hybridization capacity of Fe_3O_4@PDA@cDNA was found to be associated with the salt concentration in the hybridization buffer. As shown in Fig. 4A, the percentage of Apt-F hybridized on Fe_3O_4@PDA@cDNA increased with increasing NaCl concentration. This can be attributed to the charge screening effect of NaCl on the electrostatic repulsion between Apt-F and immobilized cDNA-NH_2 as well as between Apt-F and the PDA surface. Notably, chloride ions quench fluorescence [29]; thus, the reference fluorescence of Apt-F should be individually measured at the corresponding NaCl concentration. The percentages of Apt-F binding to Fe_3O_4@PDA and Fe_3O_4@PDA@ncDNA (noncomplementary DNA) were significantly lower (Fig. 4A), indicating minimal nonspecific adsorption and excellent sequence selectivity. However, when the NaCl concentration exceeded 415 mM, nonspecific adsorption started to increase. We found that nonspecific adsorption at high NaCl concentrations could be effectively mitigated by adding 0.02% Tween 20 to the hybridization buffer (Fig. 4B).

The hybridization kinetics were monitored by incubating Fe_3O_4@PDA@cDNA (0.1 mg mL^{-1}) with Apt-F (50 nM) for 6–90 min (Fig. 4C). Hybridization occurred rapidly, and approximately half of the added Apt-F was captured within 10 min. The maximum hybridization capacities of the Fe_3O_4@PDA@cDNA conjugates prepared with 0 or 20 mM Mg^{2+} were measured and compared (Fig. 4D). Obviously, the conjugates prepared with 20 mM Mg^{2+} had a greater hybridization capacity. For 0.1 mg mL^{-1} of Fe_3O_4@PDA@cDNA, up to ~65 nM Apt-F was hybridized to conjugates prepared with 20 mM Mg^{2+}, while only ~19 nM was hybridized to the conjugates prepared without Mg^{2+}.

The chemical stability of the linkage between cDNA-NH_2 and Fe_3O_4@PDA was evaluated by examining the hybridization capacity of Fe_3O_4@PDA@cDNA with Apt-F after storage at 4 °C for 1–10 days. As shown in Fig. 4E, no significant decrease in the hybridization percentage was observed, indicating the excellent chemical stability of the conjugates. Five batches of the Fe_3O_4@PDA@cDNA conjugates were prepared under the optimized conditions. As shown in Fig. 4F, the hybridization capacities of these conjugates with Apt-F were comparable, indicating excellent reproducibility of the conjugation reaction.

The Fe_3O_4@PDA@cDNA conjugates hybridized with Apt-F can be regenerated by denaturing the double strands at high temperature. Initially, hybridization was performed at high concentrations of NaCl to achieve high hybridization efficiency. Thereafter, the Fe_3O_4@PDA@cDNA/Apt-F complexes were redispersed in a low-ionic-strength solution (10 mM Tris-HCl, pH 8.5) and regenerated by heating at 50 °C for 10 min. The reusability was assessed by repeating the hybridization–regeneration cycle 5 times. The hybridization and dissociation percentages are presented in Fig. 4G; both were calculated relative to the total Apt-F. No reduction in hybridization capacity occurred after regeneration, indicating the excellent reusability of the

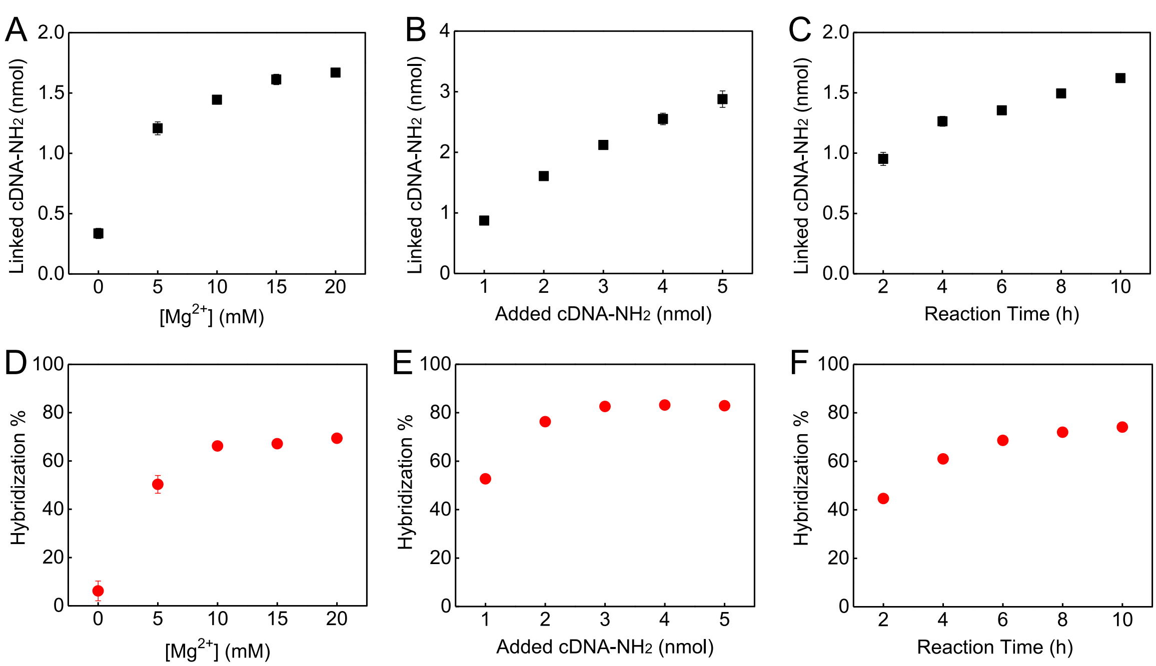

Fig. 3. Optimization of the reaction conditions for the conjugation of cDNA-NH_2 on Fe_3O_4@PDA nanoparticles: (A, D) Mg^{2+} concentration, (B, E) cDNA-NH_2 dose, and (C, F) reaction time. (A–C) The amount of cDNA-NH_2 covalently linked on the Fe_3O_4@PDA nanoparticles under each tested condition. (D–F) The hybridization percentages of Apt-F on the prepared Fe_3O_4@PDA@cDNA conjugates. n = 3.

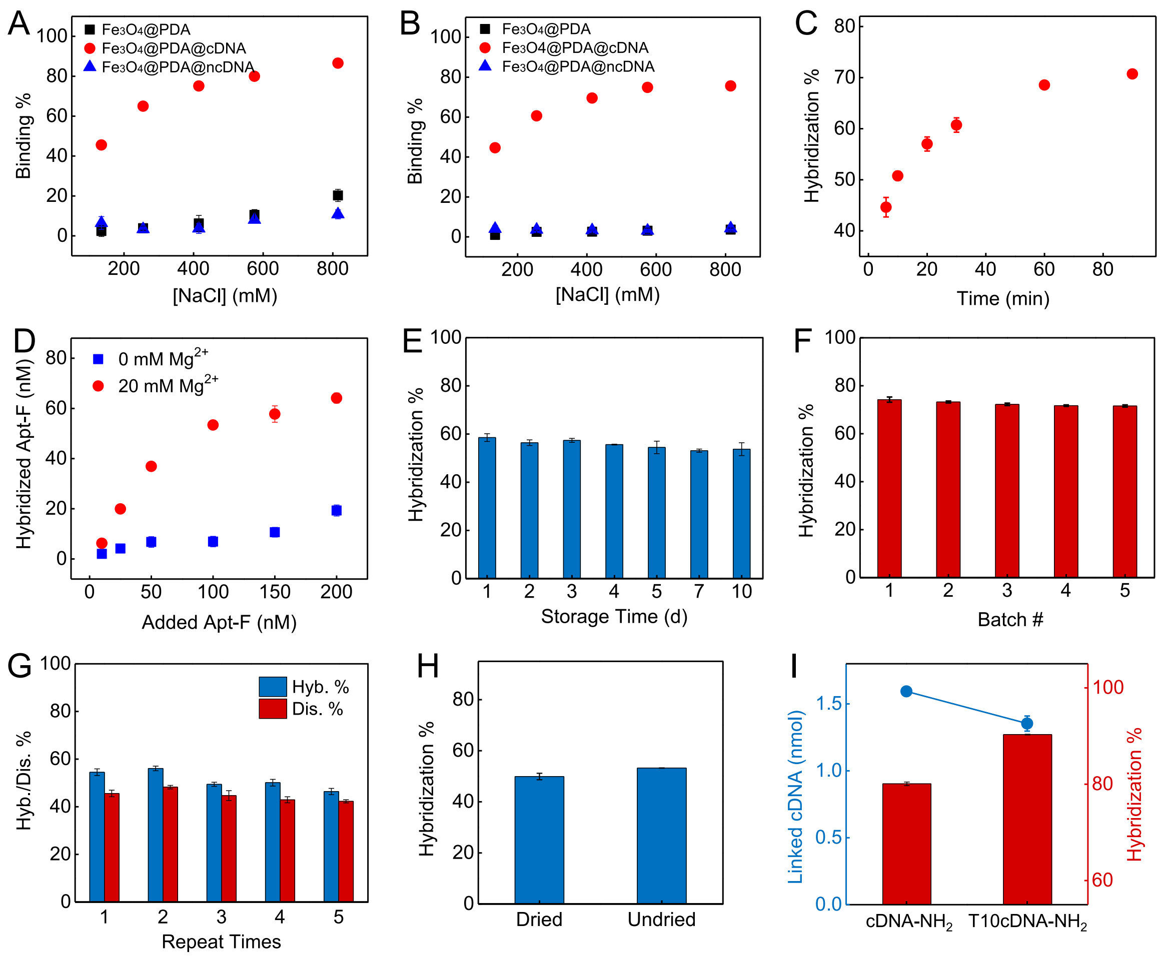

Fig. 4. Key properties of the Fe_3O_4@PDA@cDNA conjugates. (A, B) The percentages of Apt-F binding (hybridization or nonspecific adsorption) to Fe_3O_4@PDA, Fe_3O_4@PDA@cDNA, and Fe_3O_4@PDA@ncDNA nanoparticles in the presence of (A) varying concentrations of NaCl and (B) varying concentrations of NaCl and 0.02% Tween 20. (C) Hybridization kinetics of Apt-F and Fe_3O_4@PDA@cDNA. (D) Comparison of the hybridization capacities of the Fe_3O_4@PDA@cDNA conjugates prepared with 0 or 20 mM Mg^{2+}. (E) Chemical stability of the linkage between cDNA-NH_2 and Fe_3O_4@PDA. (F) Reproducibility of the conjugation reaction. (G) Reusability of the Fe_3O_4@PDA@cDNA conjugates. The hybridization percentages (Hyb. %) and dissociation percentages (Dis. %) were both calculated relative to the total Apt-F. (H) Comparison of the hybridization capacities of vacuum-dried and undried Fe_3O_4@PDA@cDNA conjugates. (I) Comparison of the amounts of cDNA-NH_2 and T10cDNA-NH_2 (with a T10 spacer) linked to Fe_3O_4@PDA and the percentages of Apt-F hybridized to the obtained Fe_3O_4@PDA@cDNA and Fe_3O_4@P-DA@T10DNA conjugates. Unless otherwise specified, the hybridization conditions were 0.1 mg mL^{-1} magnetic nanoparticles and 50 nM Apt-F in hybridization buffer (10 mM Tris-HCl at pH 8.5) supplemented with 800 mM NaCl and 0.02% Tween 20 for an incubation time of 20 min (for E, G, and H) or 60 min (for A-D, F, and I). n = 3.

Fe_3O_4@PDA@cDNA conjugates.

Additionally, we tested the hybridization capacity of vacuum-dried Fe_3O_4@PDA@cDNA and found that it maintained a hybridization capacity comparable to that of the undried conjugates (Fig. 4H). The vacuum-dried conjugates could be rapidly redispersed via ultrasonication and exhibited excellent aqueous dispersibility.

For the immobilization of DNA, a spacer is commonly employed to reduce steric hindrance and electrostatic interactions from adjacent DNA strands and increase accessibility to the target [30,31]. Herein, we inserted a T10 spacer between cDNA and the terminal amine, referred to as T10cDNA-NH_2, which was employed to functionalize Fe_3O_4@PDA nanoparticles. The introduction of the T10 spacer resulted in a decrease in linked amounts from ~1.59 nmol to ~1.35 nmol, but an increase in hybridization percentages with Apt-F from 80% to 90% (Fig. 4I). Therefore, the Fe_3O_4@PDA nanoparticles modified with T10cDNA-NH_2 were used for subsequent OTA detection.

Overall, the Fe_3O_4@PDA@cDNA conjugates prepared via the Mg^{2+}-mediated approach exhibited excellent properties, including enhanced hybridization capacity, rapid hybridization kinetics, minimal nonspecific adsorption, superior chemical stability, reusability, and so on. These features make them promising candidates for a wide range of applications in target extraction, enrichment, and sensing.

3.5. Aptamer-based OTA detection

Ochratoxin A (OTA) is a ubiquitous mycotoxin that exists in food products and has been classified as a possible human carcinogen (Group 2B) [32]. Aptamers are short single-stranded nucleic acid sequences with high affinity for specific targets. Compared to antibodies, aptamers have the advantages of facile synthesis and modification, low batch-to-batch variation and high stability [33]. Herein, we used aptamer-based OTA detection as an example to demonstrate the applicability of the Fe_3O_4@PDA@T10cDNA conjugates synthesized by the Mg^{2+}-mediated method. In detail, FAM-labelled OTA aptamers (Apt-F) were first mixed with OTA. A portion of Apt-F was combined with OTA, while the remaining unbound Apt-F molecules were captured by the Fe_3O_4@PDA@T10cDNA conjugates and removed via magnetic separation. The fluorescence intensities of the supernatants exhibited a positive correlation with OTA concentrations. A schematic diagram of this assay process is shown in Fig. 5A.

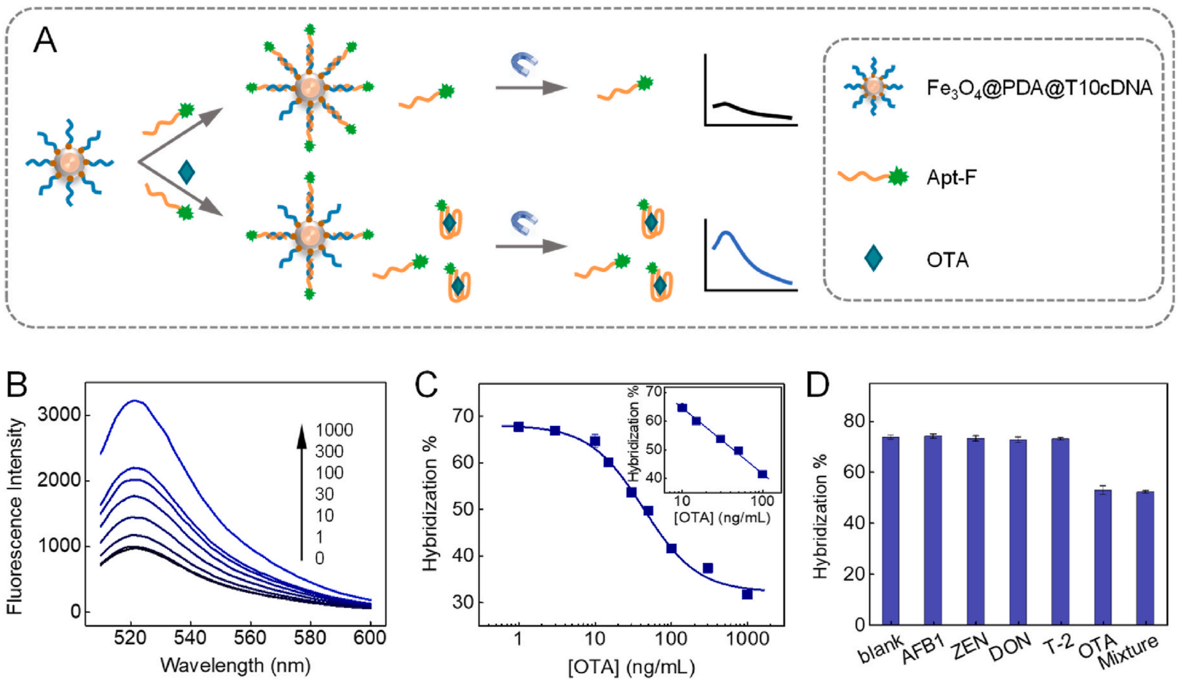

Fig. 5. (A) A schematic diagram of the aptamer-based OTA detection method. (B) Fluorescence spectra of supernatants in the presence of various concentrations of OTA; the fluorescence spectrum of Apt-F free of hybridization was used as a control. (C) A semilogarithmic plot of the hybridization percentages of Apt-F verses the OTA concentrations, exhibiting a linear relationship in the range of 10–100 ng mL^{-1} OTA (inset). (D) Selectivity of the OTA detection method. n = 3.

The detection conditions were optimized. As demonstrated above, the concentration of NaCl impacts the hybridization between Apt-F and Fe_3O_4@PDA@T10cDNA. In addition, Mg^{2+} or Ca^{2+} are necessary for aptamer–OTA recognition [34]. Thus, the concentrations of NaCl, $MgCl_2$, and $CaCl_2$ in the binding buffer were first finely tuned to achieve increased sensitivity for OTA detection. Fig. S4 presents the hybridization percentages of Apt-F on Fe_3O_4@PDA@T10cDNA in the presence and absence of 100 ng mL^{-1} OTA as well as their differences under varying concentrations of NaCl, $MgCl_2$, and $CaCl_2$. A higher difference in hybridization percentage indicates greater sensitivity for OTA detection. The results demonstrated that NaCl facilitated the capture of unbound Apt-F by Fe_3O_4@PDA@T10cDNA but had little effect on sensitivity. However, increasing the concentrations of $MgCl_2$ or $CaCl_2$ significantly improved the sensitivity, and $CaCl_2$ yielded a greater difference in hybridization percentage than $MgCl_2$. Although polyvalent metal ions can promote the adsorption of Apt-F on Fe_3O_4@PDA@T10cDNA, this process can be effectively inhibited by adding Tween 20 to the binding buffer (Fig. S5). Nevertheless, when the concentration of Ca^{2+} exceeded 4 mM, polyvalent-metal-mediated adsorption began to increase. Finally, a binding buffer containing final concentrations of 600 mM NaCl and 3.6 mM $CaCl_2$ was used for OTA detection.

In addition, the length of the complementary region in T10cDNA-NH_2 was optimized. T10cDNA-NH_2 molecules with 10–15 nt of complementary regions were tested (Fig. S6A). The length of the complementary region had little effect on the sensitivity. T10cDNA-NH_2 with a 15 nt of complementary region was ultimately selected. Keeping the concentration of Apt-F at 50 nM, the concentration of Fe_3O_4@PDA@T10cDNA conjugates was regulated, and a final concentration of 0.15 mg mL^{-1} was selected (Fig. S6B).

Under the optimized conditions, OTA was detected. The fluorescence spectra of the supernatants in the presence of various concentrations of OTA are presented in Fig. 5B, and the fluorescence intensities increased with increasing OTA concentration. A calibration curve depicting the hybridization percentages of Apt-F verses the logarithm of OTA concentrations is shown in Fig. 5C, with a linear range of 10–100 ng mL^{-1} (R^2 = 0.996). The limit of detection (LOD) based on 3 times the signal-to-noise ratio (3S/N) was determined to be 2.77 ng mL^{-1}.

To evaluate the selectivity of the OTA detection method, four other common mycotoxins, AFB1, ZEN, DON, and T-2, as well as OTA, were analysed at a concentration of 100 ng mL^{-1}. As shown in Fig. 5D, only OTA caused an obvious decrease in hybridization percentage, indicating good selectivity for OTA detection. Moreover, the presence of other mycotoxins did not interfere with the response of OTA.

To demonstrate the feasibility of this OTA detection method for real sample assays, red wines spiked with 25, 50, 100, or 200 ng mL^{-1} OTA were analysed with this aptamer-based method and verified by ELISA (Table 1). These results confirmed the potential applicability of this method for OTA determination in real samples.

3.6. Versatility

Mg^{2+} was successfully used to promote the conjugation of cDNA-NH_2 to Fe_3O_4@PDA nanoparticles through polyvalent metal-mediated DNA adsorption on PDA surfaces. To extend this method, another divalent metal ion, Ca^{2+}, was tested. cDNA-NH_2 was conjugated to Fe_3O_4@PDA in the presence of 0–10 mM Ca^{2+}, and the amount of covalently linked cDNA-NH_2 at each tested Ca^{2+} concentration is plotted in Fig. 6A. Ca^{2+} could also facilitate the conjugation of cDNA-NH_2 to Fe_3O_4@PDA. Moreover, a lower Ca^{2+} concentration was needed to achieve an equivalent level of conjugation compared to that of Mg^{2+}. The

Table 1
Determination of OTA in spiked red wine samples.

Added (ng/mL)	Aptamer-based method			ELISA method		
	Found (ng/mL)	Recovery (%)	RSD (%)	Found (ng/mL)	Recovery (%)	RSD (%)
25	26.5 ± 3.07	106.0	11.5	23.8 ± 1.79	95.4	7.49
50	53.6 ± 2.98	107.2	5.56	49.5 ± 1.38	99.0	2.78
100	97.2 ± 3.30	97.2	3.39	97.2 ± 12.3	97.2	12.6
200	200.8 ± 2.82	100.4	1.41	188.6 ± 4.98	94.3	2.64

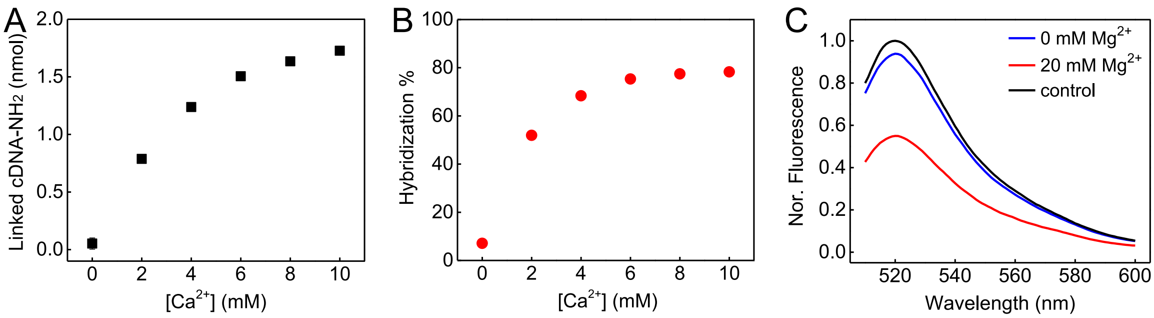

Fig. 6. (A) Covalently linked amounts of cDNA-NH_2 on Fe_3O_4@PDA nanoparticles promoted by 0–10 mM Ca^{2+}. (B) The hybridization percentages of Apt-F on the obtained Fe_3O_4@PDA@cDNA conjugates in (A). (C) Normalized fluorescence spectra of Apt-F after incubation in PDA-coated wells modified with cDNA-NH_2 in the presence or absence of 20 mM Mg^{2+}, taking Apt-F without incubation as a control.

hybridization capacities of the obtained Fe_3O_4@PDA@cDNA conjugates with Apt-F are presented in Fig. 6B. ~7% and ~78% of the Apt-F were hybridized on the Fe_3O_4@PDA@cDNA conjugates prepared with 0 mM and 10 mM Ca^{2+}, respectively.

Since PDA can be deposited on nearly any material surface, such as in 96-well plates, the conjugation of cDNA-NH_2 to a PDA-coated 96-well plate was promoted by Mg^{2+}. Fig. 6C shows a comparison of the hybridization capacities of cDNA-NH_2 linked to PDA-coated wells with 0 or 20 mM Mg^{2+}, with Apt-F free of hybridization serving as a control. Similarly, the wells modified with cDNA-NH_2 in the presence of 20 mM Mg^{2+} hybridized more Apt-F than the wells modified with cDNA-NH_2 in the absence of Mg^{2+}, indicating that Mg^{2+} promoted the conjugation of cDNA-NH_2 on PDA-coated 96-well plates.

4. Conclusions

In summary, in this work we provided a facile and versatile method for high-efficiency DNA attachment. When promoted by polyvalent metal ions (Mg^{2+} or Ca^{2+}), the conjugation efficiency of cDNA-NH_2 on PDA-coated substrates (Fe_3O_4@PDA nanoparticles and PDA-coated 96-well plates) was significantly enhanced. After the reaction, the unlinked cDNA-NH_2 molecules were effectively removed via EDTA chelation. The obtained Fe_3O_4@PDA@cDNA conjugates exhibited a remarkable improvement in hybridization capacity, minimal nonspecific adsorption, high stability, and reusability. These features make them highly suitable for a wide range of applications in sequence-specific DNA extraction and biosensing. For example, the conjugates were employed in aptamer-based OTA detection, achieving an LOD of 2.77 ng mL^{-1}.

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CRediT authorship contribution statement

Wei Yang: Conceptualization, Data curation, Investigation, Methodology, Validation, Visualization, Writing – original draft. **Liang Feng:** Conceptualization, Funding acquisition, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.aca.2024.342382.

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